



ME 300: Engineering Thermodynamics

Energy Properties

Internal Energy

→ energy associated with the motion of molecules

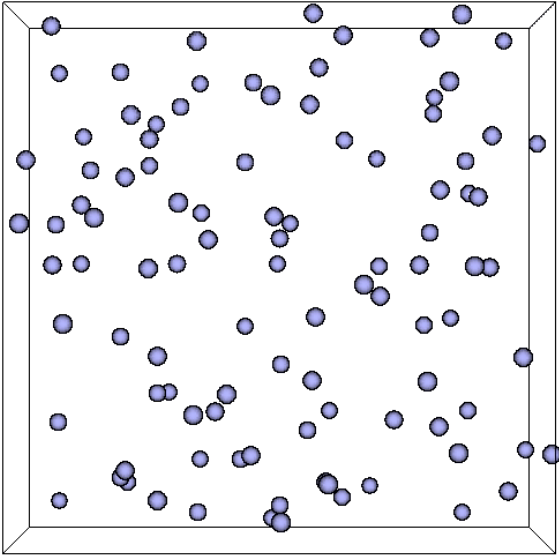
→ translation energy → $U_{\text{trans}} = \frac{1}{2} N^{\text{m}} \bar{v}^2$ ← molecular speed
 $\uparrow$
 $\left[\frac{\text{kg}}{\text{mol}} \right]$

→ vibrate

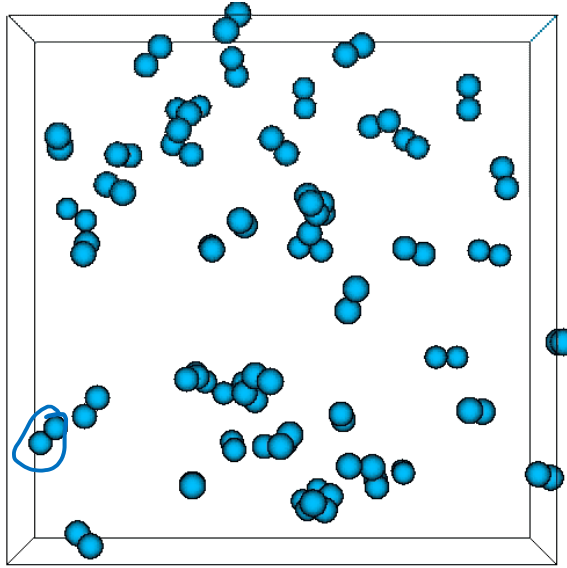
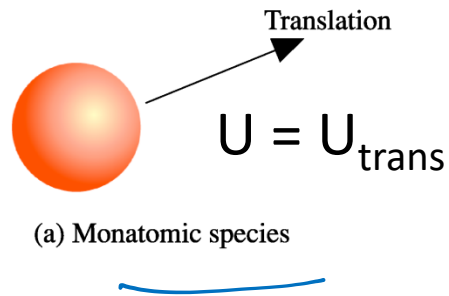
→ rotate

Demonstration

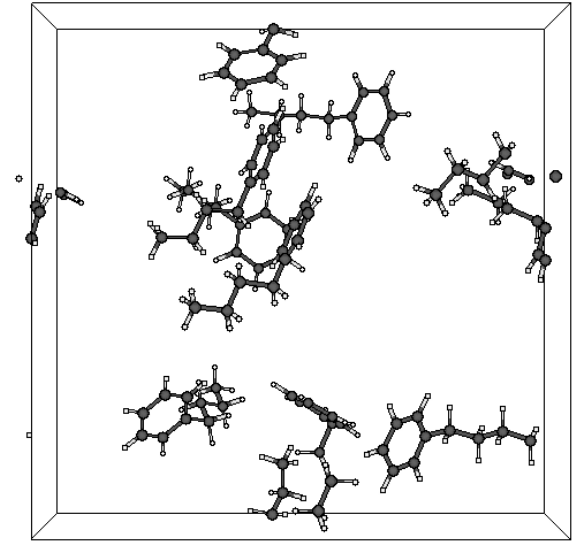
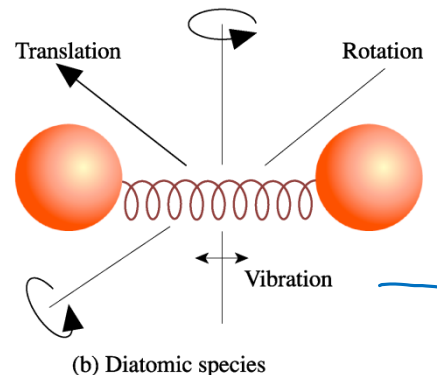
Internal Energy (videos by Adri van Duin)



He-gas at $T=500\text{K}$
Translational energy only



N₂-gas at $T=500\text{K}$:
translational + rotational
+ bond-stretch vibration



Butylbenzene at
 $T=500\text{K}$: complex
internal molecular
motion

$$U = U_{\text{trans}} + U_{\text{rot}} + U_{\text{vib}}$$

Enthalpy

$$H = U + PV \sim [J]$$

$$h = u + Pv \sim [J/kg]$$

$$\bar{h} = \bar{u} + P\bar{v} \sim [J/mol]$$

Specific heat

→ capacity of a substance to store energy

$$\rightarrow c_v = \left(\frac{\partial u}{\partial T} \right)_v \sim \left[\frac{\text{J}}{\text{kg} \cdot \text{K}} \right] \quad \bar{c}_v = \left(\frac{\partial \bar{u}}{\partial T} \right)_v \sim \left[\frac{\text{J}}{\text{mol} \cdot \text{K}} \right]$$

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p \sim \left[\frac{\text{J}}{\text{kg} \cdot \text{K}} \right] \quad \bar{c}_p = \left(\frac{\partial \bar{h}}{\partial T} \right)_p \sim \left[\frac{\text{J}}{\text{mol} \cdot \text{K}} \right]$$